

# V1 TKA Preoperative Planning Software

## Detailed Software UI Workflow & Click-by-Click User Interaction Specification

**Version:** 1.0.0 (Production V1 Specification)  
**System Architecture:** Preoperative 2D Radiographic Planner → VR Execution Pipeline  
**Target Users:** Orthopedic Surgeons, Preoperative Surgical Planners

### 1. Executive Core Scope & Principles

This document specifies the exact click-level software workflow for the V1 Total Knee Arthroplasty (TKA) Preoperative Planning Software. The mental model of V1 is strictly governed by three verbs: **MEASURE** → **SIZE** → **SEND**. No intraoperative variables, 3D bone cut depths, resection planes, soft-tissue balancing, or alignment optimizations are present in this version. All intraoperative surgical decisions are explicitly deferred to the downstream VR surgical environment.

#### ■ V1 MANDATORY ARCHITECTURAL RULES

- Strict 4-Page Linear Workflow: Assessment → Tibial Planning → Femoral Planning → Review & Send to VR.
- Deterministic State Progression: Unprocessed steps are disabled. Navigation buttons remain locked until strict clinical completion thresholds are met.
- Radio-opaque Calibration Required: Calibration markers are automatically detected. Implant overlays scale dynamically based on pixels-per-millimeter (px/mm) patient scale.
- Zero Extraneous Complexity: Only 6 structural anatomical measurements for assessment, and 4 for component sizing measurements for positioning of implants, are executed - A total of 10 measurement steps.

### 2. Global UI Layout & Persistent Patient Banner

Across all four primary pages, the top of the interface displays a persistent, non-scrollable Global Header Bar containing critical patient metadata, active view status, and hard locks.

| UI Header Element | Data Field / Indicator        | Source                               | Interactive Behavior             |
|-------------------|-------------------------------|--------------------------------------|----------------------------------|
| Patient Identity  | Patient ID, Age, Sex          | DICOM Metadata (Hardcoded for V1)    | Read-only.                       |
| Surgical Side     | Operative Knee (Left / Right) | DICOM / Selection (Hardcoded for V1) | Hard-locked after session start. |
| Procedure Type    | Primary TKA                   | System Default (Case dependent)      | Read-only.                       |

| UI Header Element | Data Field / Indicator | Source       | Interactive Behavior                   |
|-------------------|------------------------|--------------|----------------------------------------|
| Plan State Status | Draft / Locked & Sent  | System State | Updates dynamically upon confirmation. |

### 3. Page-by-Page Click-by-Click Interaction Model

#### PAGE 1: ASSESSMENT (Radiographic Measurement)

##### Page Objective

Establish the baseline anatomical axes and joint geometry from calibrated 2D radiographs. Complete exactly six required measurements across two radiographic views.

##### Step 0: DICOM View Selection & Viewport Layout

The screen presents a two-panel layout: Left Sidebar (Measurement Control Panel) and Center/Right Viewport (Radiograph Display). At the top of the Left Panel, two segmented toggle tabs control the active radiograph view:

- 1. [● FLAP] — Full-Length Standing Anteroposterior Radiograph (Default active view).
- 2. [○ KLAT] — Knee Lateral Radiograph.

##### Step 1: Executing FLAP Measurements (5 Measurements)

The Left Sidebar displays five measurement cards. Each card displays an unfulfilled icon (○), the measurement abbreviation, full title, and status text ('Not Measured').

| Measurement | View | Clicks / Points Required                                                                                                            | Calculation / Definition                                                                                                                    |
|-------------|------|-------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------|
| MAD         | FLAP | 1. Femoral Head Center<br>2. Knee Joint Center<br>3. Ankle Joint Center                                                             | Perpendicular distance (mm) from the knee joint center to the mechanical axis (femoral head → ankle center).                                |
| AMA         | FLAP | 1. Proximal Femoral Shaft Point<br>2. Distal Femoral Shaft Point                                                                    | Angle (°) between the femoral anatomical axis defined by the two shaft points and the femoral mechanical axis (femoral head → knee center). |
| mHKA        | FLAP | 1. Femoral Head Center<br>2. Knee Joint Center<br>3. Ankle Joint Center                                                             | Included angle (°) between the femoral mechanical axis (H → K) and tibial mechanical axis (K → A). Display with Varus / Valgus designation. |
| MPTA        | FLAP | 1. Medial Tibial Plateau Point<br>2. Lateral Tibial Plateau Point<br>3. Proximal Tibial Shaft Point<br>4. Distal Tibial Shaft Point | Medial angle (°) between the tibial plateau/joint line and the tibial mechanical axis defined by the two shaft points.                      |

| Measurement | View | Clicks / Points Required                                       | Calculation / Definition                                                                                              |
|-------------|------|----------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------|
| LDFA        | FLAP | 1. Medial Distal Femoral Point 2. Lateral Distal Femoral Point | Lateral angle (°) between the distal femoral joint line and the femoral mechanical axis (femoral head → knee center). |

**\*Anatomical landmarks can be reused once placed. So, no unwanted repeated placement should be there for users.**

#### **Detailed Click & Calculation Flow per Measurement:**

1. User Clicks Measurement Card (e.g., [o MAD]). Card highlights in Steel Blue border.
2. *On-Screen Guidance Prompt appears at top of viewport: 'Place Point 1: Center of Femoral Head.'*
3. *User Clicks Target Point on Viewport. A vector node appears with magnifying lens hover assistance. Prompt updates: 'Place Point 2: Center of Ankle Talus.'*
4. User Clicks Second Point. System draws vector line, computes mathematical geometry instantly against the calibrated DICOM pixel grid, and populates value.
5. Card UI Updates: Icon transitions to green checkmark (✓ MAD), value is displayed in bold (e.g., '12 mm Varus'), and 'Adjust Points' option unlocks.

#### **Step 2: Switching View & Executing KLAT Measurement**

1. User Clicks Tab: [● KLAT]. Viewport smoothly transitions to the Knee Lateral radiograph.
2. *Left Sidebar automatically updates list to present remaining assessment measurement: [o PTS] Posterior Tibial Slope.*
3. *User Clicks [o PTS]. Prompt instructs: 'Place 2 points along proximal tibial anterior cortex, then 1 point along tibial plateau tangent line.'*
4. *User Clicks Points. System calculates slope relative to longitudinal shaft line. Output populates: ✓ PTS 7°.*

#### **Step 3: Assessment Page Completion & Transition**

Once all 6 measurements display green checkmarks (✓), the primary action button at the bottom right corner transitions from Grey Disabled to Active Blue:

[ Continue to Tibial Planning → ]

Clicking this button hard-locks Assessment values and loads Page 2.

PAGE 2: TIBIAL PLANNING (Sizing & 2D Overlay Fit)

| Measurement         | View | Used for                | What is measured                                   |
|---------------------|------|-------------------------|----------------------------------------------------|
| Tibial AP Dimension | KLAT | Tibial component sizing | Anterior–posterior dimension of the proximal tibia |
| Tibial ML Dimension | FLAP | Tibial component sizing | Medial–lateral dimension of the proximal tibia     |

Page Objective

Determine the optimal tibial component size and verify 2D anatomical fit (coverage and overhang) using calibrated CAD template overlays on the AP/Lateral radiograph.

Step 1: Automated Initial Sizing Input

The left panel displays the **Tibial Sizing Section**. User computes AP and ML dimensions on the Xray.

- Radiographic AP Dimension: 42.5 mm
- Radiographic ML Dimension: 68.2 mm

Based on these numbers, the system auto-highlights the best-matching implant size button in the size selection toolbar.

Step 2: Implant Size Selection Toolbar

The user is presented with a discrete horizontal toggle list representing available manufacturer sizes:

[ Size 1 ] [ Size 2 ] [ Size 3 - Suggested ] [ Size 4 ] [ Size 5 ] [ Size 6 ]

How the user selects: Clicking any size button immediately swaps the active 2D CAD template overlay rendered in the viewport.

Step 3: Interactive 2D Implant Positioning

- 1. **CAD Template Rendering:** A semi-transparent blue vector graphic of the selected tibial tray size overlays the proximal tibia in the AP viewport.
- 2. **Mouse Interaction:** The user clicks and drags the center handle of the template to translate it in 2D space (Medial/Lateral, Anterior/Posterior on KLAT tab). Rotational handle allows minor 2D alignment with MPTA axis line.

Step 4: Real-Time Dynamic Fit & Overhang Calculation

As the user drags or changes component size, the software performs real-time polygon intersection between the implant outer geometry boundary and the annotated bone border. The 'FIT METRICS' panel updates live:

| Fit Metric Name   | Real-Time Value | Acceptable Threshold | Visual UI Status Indicator |
|-------------------|-----------------|----------------------|----------------------------|
| Cortical Coverage | 96.4%           | ≥ 90.0%              | Green Text / Checkmark     |

| Fit Metric Name    | Real-Time Value | Acceptable Threshold | Visual UI Status Indicator              |
|--------------------|-----------------|----------------------|-----------------------------------------|
| Medial Overhang    | 0.2 mm          | ≤ 1.0 mm             | Green Text / Checkmark                  |
| Lateral Overhang   | 0.0 mm          | ≤ 1.0 mm             | Green Text / Checkmark                  |
| Overall Fit Status | ACCEPTABLE FIT  | All thresholds met   | Green Status Badge [ ✓ FIT ACCEPTABLE ] |

If overhang exceeds 1.5 mm, the border turns Amber/Red with a warning tag: 'CAUTION: Medial Overhang > 1.5mm'.

Step 5: Lock Tibial Plan

User clicks [ Confirm Tibial Component ]. System saves: Selected Size (e.g., Size 3), 2D X/Y Offset Coordinates. Interface enables navigation button: [ Continue to Femoral Planning → ]

PAGE 3: FEMORAL PLANNING (Sizing & 2D Overlay Fit)

| Measurement          | View | Used for                 | What is measured                                 |
|----------------------|------|--------------------------|--------------------------------------------------|
| Femoral AP Dimension | KLAT | Femoral component sizing | Anterior–posterior dimension of the distal femur |
| Femoral ML Dimension | FLAP | Femoral component sizing | Medial–lateral dimension of the distal femur     |

Page Objective

Select femoral component size and verify anatomical fit, re-using the exact same interaction model as Page 2 to ensure cognitive consistency.

Step-by-Step Execution Sequence

- 1. Radiographic Dimensions Displayed: System displays measured Femoral AP (58.4 mm) and ML (64.1 mm).
- 2. Size Selection: User selects size from horizontal selector [ Size 1 ] through [ Size 8 ]. Size 4 auto-suggested.
- 3. 2D Drag & Overlay: User overlays femoral template over distal femur. User toggles to KLAT view to verify anterior condylar flush line and check for anterior notching.
- 4. Dynamic Fit Calculation Updates:
  - AP Coverage: 97.1%
  - ML Coverage: 95.8%
  - Anterior Notching Risk: 0.0 mm (Flush)
  - Fit Status: [ ✓ FIT ACCEPTABLE ]
- 5. Lock Femoral Plan: User clicks [ Confirm Femoral Component ]. Software unlocks button: [ Continue to Review → ]

## PAGE 4: REVIEW & SEND TO VR (Plan Lock & Export)

### Page Objective

Provide a clean, immutable read-only summary of all completed assessment metrics and selected implants prior to locking and transmitting data to the VR simulation suite.

### Summary Screen Layout

The screen displays three structured cards side-by-side:

| Card 1: Patient & Radiographs                                                                     | Card 2: Assessment Measurements                                                                                                                                                                                                       | Card 3: Component Selections                                                                                                                                                                                                                                                                     |
|---------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Patient ID: P-0247<br>Operative Knee: Right<br>Procedure: Primary TKA<br>Calibration: 0.264 mm/px | FLAP: <ul style="list-style-type: none"><li>• MAD: 12 mm Varus</li><li>• AMA: 6.0°</li><li>• mHKA: 7.0° Varus</li><li>• MPTA: 89.0°</li><li>• LDFA: 88.0°</li></ul> KLAT: <ul style="list-style-type: none"><li>• PTS: 7.0°</li></ul> | Tibial Component: <ul style="list-style-type: none"><li>• Selected Size: 3</li><li>• Fit Status: Acceptable</li><li>• Coverage: 96.4%</li></ul> Femoral Component: <ul style="list-style-type: none"><li>• Selected Size: 4</li><li>• Fit Status: Acceptable</li><li>• Coverage: 97.1%</li></ul> |

### Final Confirmation & Transmission Actions

Bottom Navigation Toolbar contains two buttons:

1. [ ← Back to Planning ]: Allows returning to Page 2 or 3 to modify component sizes if needed.
2. [ LOCK PLAN & SEND TO VR → ]: Primary action button (Highlighted Solid Navy Blue).

On Clicking [ LOCK PLAN & SEND TO VR → ]:

- a. Modal confirmation pops up: 'Lock Preoperative Plan? Once locked, parameters cannot be modified in 2D software.'
- b. User clicks [ Confirm & Send ].
- c. System serializes patient data, measurements, and implant sizes into standard VR JSON payload.
- d. UI displays success toast: 'Plan successfully transmitted to VR Suite. Session Completed.'

## 4. Data Transfer Payload Specification (Output to VR)

To ensure complete technical clarity, the exact data structure transmitted from the 2D planning software to the VR surgical environment is detailed below. Non-V1 parameters (such as bone cuts or gap balancing) are strictly omitted.

### ■ JSON PAYLOAD STRUCTURE SENT TO VR

- {  
  "patient\_id": "P-0247",

```
"knee_side": "RIGHT",
"assessment": {
  "MAD_mm": 12.0,
  "AMA_deg": 6.0,
  "mHKA_deg": 7.0,
  "MPTA_deg": 89.0,
  "LDFA_deg": 88.0,
  "PTS_deg": 7.0
},
"tibial_component": {
  "implant_size": 3,
  "position_2d": { "x_offset_mm": 1.2, "y_offset_mm": -0.4, "rotation_deg": 0.5 }
},
"femoral_component": {
  "implant_size": 4,
  "position_2d": { "x_offset_mm": 0.5, "y_offset_mm": 0.0, "rotation_deg": 0.0 }
}
}
```